CREDIT RISK ANALYSIS SYSTEM

Data Science and Machine Learning Project

Presented By: Maha

Project Domain: Data Science

Tools: Python, Pandas, NumPy, Scikit-learn

PROJECT INTRODUCTION :

Credit Risk Analysis is the process of evaluating a customer's ability to repay a loan.

This project uses Data Science and Machine Learning techniques to:

• Analyze customer financial data.

• Predict loan approval status.

• Identify customer credit risk levels.

• Generate loan recommendations.

• Support better risk assessment.

PROJECT OBJECTIVES :

- To collect and preprocess customer financial data.
- To perform Exploratory Data Analysis (EDA).
- To identify important factors affecting credit risk.
- To build Machine Learning models for loan prediction.
- To classify customers based on credit risk.
- To generate final loan recommendations.

PROJECT TECHNOLOGIES :

PROGRAMMING LANGUAGE:

- Python

DATA ANALYSIS:

- Pandas
- NumPy

DATA VISUALIZATION:

- Matplotlib
- Seaborn

MACHINE LEARNING:

- Scikit-learn

DEVELOPMENT PLATFORM:

- Google Colab

PROJECT MODULES :

MODULE 1 – DATA COLLECTION AND PREPROCESSING

MODULE 2 – EXPLORATORY DATA ANALYSIS (EDA)

MODULE 3 – CREDIT RISK PREDICTION MODEL

MODULE 4 – RISK SCORE DASHBOARD

MODULE 5 – FINAL RECOMMENDATION SYSTEM

DATA COLLECTION AND PREPROCESSING :

- Imported the dataset into Python using Pandas.
- Inspected the dataset rows, columns, and data types.
- Identified numerical and categorical columns.
- Checked dataset quality and prepared features for analysis.
- Prepared loan status as the prediction target.

EXPLORATORY DATA ANALYSIS (EDA) :

- Analyzed the distribution of loan status.
- Studied numerical and categorical features.
- Compared applicant income with loan status.
- Analyzed loan amount patterns.
- Used charts and visualizations to understand data patterns.
- Identified important factors related to loan approval.

MACHINE LEARNING MODEL DEVELOPMENT :

• Prepared input features and target variable.

• Split the dataset into training and testing sets.

• Developed three machine learning models:

- Logistic Regression
- Decision Tree
- Random Forest

• Trained the models using training data.

• Evaluated model performance using test data.

MODEL COMPARISON AND SELECTION :

- Compared the performance of three machine learning models.
- Evaluated model accuracy using test data.
- Logistic Regression was included in the comparison.
- Decision Tree was included in the comparison.
- Random Forest was included in the comparison.
- Selected the best-performing model based on accuracy.

RISK SCORE DASHBOARD :

- Created a dashboard using Python.
- Displayed predicted loan approval status.
- Calculated approval probability using the selected model.
- Classified applicants into risk categories.
- Presented prediction results in a structured format.
- Saved dashboard results as a CSV file.

FINAL RECOMMENDATION SYSTEM :

- Generated loan recommendations based on model predictions.
- Approval Probability 70% or above – Approve Loan.
- Approval Probability 40% to below 70% – Manual Review.
- Approval Probability below 40% – Reject Application.
- Saved final recommendations in a CSV file.
- These thresholds are illustrative rules for this academic project.

CONCLUSION :

- Developed a Credit Risk Analysis System using Python and Machine Learning.
- Analyzed applicant data and identified important patterns.
- Built and compared multiple machine learning models.
- Created a dashboard to display prediction results.
- Generated recommendations based on approval probability.
- This project demonstrates how machine learning can support loan risk assessment.

THANKYOU!

Thank You for Your Attention!

CREDIT RISK ANALYSIS SYSTEM

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Questions & Answers